

Disentangling spatial gradients from temporal dynamics in East African ecological quality: a component-robust attribution framework

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Abstract

Satellite records show widespread vegetation change, but apparent greening can be difficult to interpret when single indicators, static environmental gradients and spatially autocorrelated validation are combined. Here we diagnose vegetation-based ecological quality across East Africa from 2001 to 2020 and show that regional improvement was concentrated in open vegetation systems where productivity-linked gains covaried most closely with climate variability. We constructed an annual ecological quality index from MODIS fractional vegetation cover, leaf area index and gross primary productivity, representing vegetation cover, canopy structure and ecosystem function. Component, weighting, threshold and leave-one-component-out tests showed that the main trend pattern was not controlled by a single component or normalization choice. False-discovery-rate-controlled trend detection identified significant improvement across 21.25% of the region and significant degradation across 3.72%, with gains concentrated mainly in grassland, savanna and shrubland contexts. Three XGBoost-SHAP models were used as model-interpretation tools to separate persistent spatial covariates, long-term trend associations and within-location annual anomalies. Soil pH and mean precipitation were the strongest spatial covariates of mean vegetation-based EQI; dewpoint, wind, temperature and precipitation trends, together with edaphic background, were associated with long-term EQI slopes; and current-year and one-year lagged precipitation anomalies were most strongly associated with annual deviations. Spatial block validation retained stable SHAP rankings, supporting a cautious but transferable framework for interpreting climate-linked vegetation-quality change in data-scarce dryland and highland mosaics.

Introduction

Vegetation change is one of the most visible satellite records of land-climate interaction, and recent global assessments show that Earth remained exceptionally green in both 2023 and 2024 [1] [2]. Yet greening is not equivalent to ecological recovery, because broad increases in vegetation indices can obscure localized mortality pulses, delayed forest recovery and short-lived degradation events [3] [4]. This distinction is central for East Africa, where grasslands, shrublands, savannas, croplands, forests, wetlands and arid margins are arranged along steep rainfall, temperature and soil gradients [5].

Remote sensing is therefore indispensable for regional monitoring, but the interpretation of land-surface change remains difficult [6]. Rapid land-change products show where surface states change, but they do not by themselves distinguish agricultural expansion, urbanization, drought, fire, restoration, woody encroachment

or mixed-pixel transitions [6]. Data-driven geospatial modelling introduces another layer of uncertainty because spatial autocorrelation, uneven sampling and predictor collinearity can make environmental relationships appear more transferable than they are [7]. Recent reviews of vegetation indices and MODIS-derived biophysical products also stress that greenness, canopy structure and productivity metrics carry different sensitivities and uncertainties [24] [25].

These uncertainties matter particularly in water-limited vegetation systems [5] [13]. African rangeland productivity is strongly associated with precipitation, but the response depends on rainfall variability, vegetation type and local environmental setting [5]. Gross primary productivity is also sensitive to wind-speed changes, optimal temperature shifts, precipitation anomalies, atmospheric dryness and soil-moisture stress [8] [10] [11] [12]. Dryland productivity can respond asymmetrically to drought, compound climate stressors and intraseasonal rainfall redistribution, which means that annual vegetation anomalies may reflect short-term climate sequencing rather than persistent ecological improvement [13] [14] [15] [18] [19]. At the same time, global GPP products and long-term GPP trends remain sensitive to model structure and climate forcing choices, so productivity-based indices need internal robustness checks [16] [17].

Land-cover context further complicates interpretation [6] [20]. Vegetation cover change is an increasingly important contributor to global leaf-area dynamics, but land-cover classes are partly coupled with the same vegetation signals used to build greenness, canopy and productivity indicators [20]. Agricultural change can alter biodiversity, water demand, energy balance and surface temperature, so cropland-associated EQI changes should be interpreted as vegetation-context changes rather than independent evidence of broader ecosystem recovery [21] [22]. Restoration and land-cover change in Africa can also modify local climate, but such biophysical effects are process-specific and cannot be inferred from vegetation EQI alone [9] [23].

We therefore treat East African vegetation-based ecological quality as a bounded diagnostic of vegetation cover, canopy structure and productivity, not as a measure of biodiversity, aquatic condition or total ecosystem health [3] [24]. The central scientific question is not whether East Africa simply became greener, but which vegetation dimensions and climate conditions are associated with the observed changes across dryland, highland and lake-influenced landscapes. We ask whether vegetation-based EQI change from 2001 to 2020 is robust to component choice and index construction, where that change accumulates across land-cover contexts, and which spatial covariates, climate-trend associations and within-location anomalies support a plausible ecological interpretation [5] [7]. We combine component consistency tests, weight and threshold sensitivity, leave-one-component-out analysis, FDR-controlled trend detection, land-cover-context interpretation and three space-time separated XGBoost-SHAP models to show that the dominant signal is productivity-linked improvement in open vegetation systems, associated with climate variability and conditioned by edaphic constraints [7] [25].

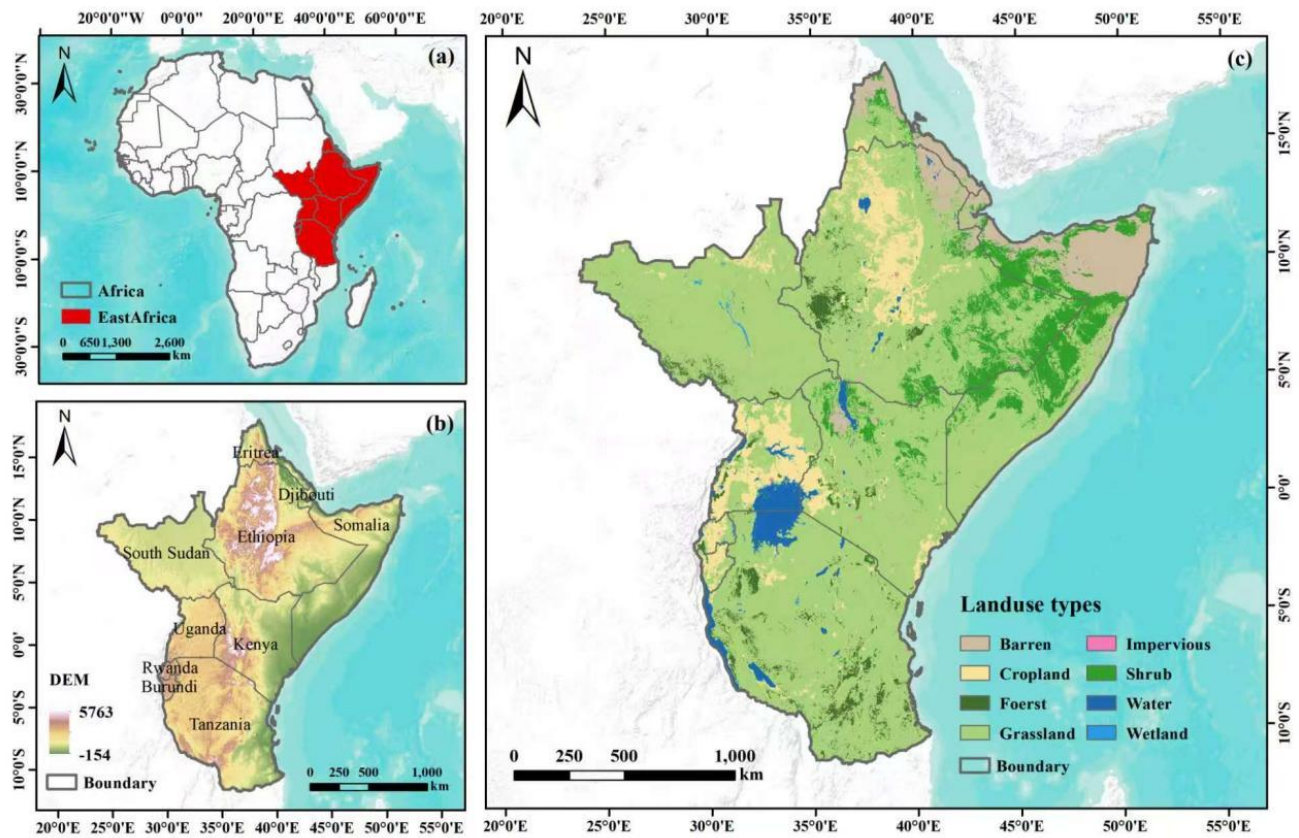


Figure 1 | Study area, elevation gradient and baseline land-cover context. a, Location of the East African study area. b, Elevation and national boundaries, highlighting the highland, Rift Valley, lake-basin and dryland gradients. c, Baseline land-cover context derived from reclassified MODIS MCD12Q1 land cover. Land cover is used to stratify ecological context, not as a standalone external explanation for the vegetation-based EQI.

Results

A component-robust vegetation index

The annual EQI was constructed from vegetation cover, canopy structure and ecosystem productivity. Component diagnostics showed that the integrated EQI was not a simple copy of any single component. Spatial Spearman correlations between mean EQI and the three components were similar in magnitude: 0.746 for FVC_{norm} , 0.754 for LAI_{norm} and 0.763 for GPP_{norm} . The components themselves were not redundant, with FVC_{norm} and LAI_{norm} strongly related but GPP_{norm} much less correlated with the structural components.

Trend consistency revealed a stronger productivity signal. GPP_{norm} had the highest slope-map agreement with EQI (Spearman $\rho = 0.846$) and the strongest overlap with significant-improvement hotspots (Jaccard = 0.687). Leave-one-component-out experiments supported this interpretation. Removing LAI_{norm} left the slope pattern nearly unchanged ($\rho = 0.984$), whereas removing GPP_{norm} reduced significant-improvement overlap to 0.472. Removing FVC_{norm} reduced degradation-hotspot overlap to 0.406, indicating that vegetation-cover loss was especially important for degradation interpretation. Across alternative weighting schemes, slope-map correlations with the default EQI were at least 0.971, and across normalization-threshold schemes that recomputed FVC_{norm} from annual NDVI they were at least 0.997 (Supplementary Tables S2 and S3).

Heterogeneous change from 2001 to 2020

The 2001-2020 mean EQI map shows a stable spatial template, with higher vegetation-based ecological quality in the Ethiopian and East African highlands, the Lake Victoria basin and other humid or lake-influenced productive regions, and lower values across the Horn of Africa drylands, northern and eastern Kenya, Eritrea, Djibouti and arid to semi-arid margins (Extended Data Fig. 1). This template reflects the vegetation-oriented

definition of EQI and should be read as a cover-structure-productivity pattern rather than as a map of total ecosystem quality.

FDR-controlled Theil-Sen and Mann-Kendall trend analysis showed spatially heterogeneous change rather than uniform regional greening. Significant improvement covered 21.25% of valid pixels, significant degradation 3.72%, non-significant improvement 32.55%, non-significant degradation 16.69% and near-stable pixels 25.79% (Fig. 2a). Improvement hotspots were concentrated mainly across extensive grassland, savanna and shrubland mosaics in parts of Kenya, Tanzania, South Sudan, Ethiopia and the Lake Victoria hinterland. Degradation was more localized and occurred mainly along dryland margins and heterogeneous agro-pastoral transition zones.

The trend classes were robust to temporal-autocorrelation sensitivity analysis. Only 7.48% of valid pixels showed significant lag-1 autocorrelation in detrended residuals. After trend-free pre-whitening and FDR correction, significant improvement covered 21.51% of the area and significant degradation 3.75%, closely matching the default result. Hotspot overlap between default and pre-whitened FDR classifications was 0.986 for significant improvement and 0.977 for significant degradation, with only 0.39% of pixels changing class.

EQI class transitions were dominated by movement among adjacent quality levels, rather than wholesale conversion from low to high quality. Low and medium-low classes persisted in 2020, but substantial flows also moved from medium and medium-low classes towards higher classes (Fig. 2b). Land-cover-specific transition ribbons show that these class shifts differed by 2001 baseline land-cover context (Fig. 3c).

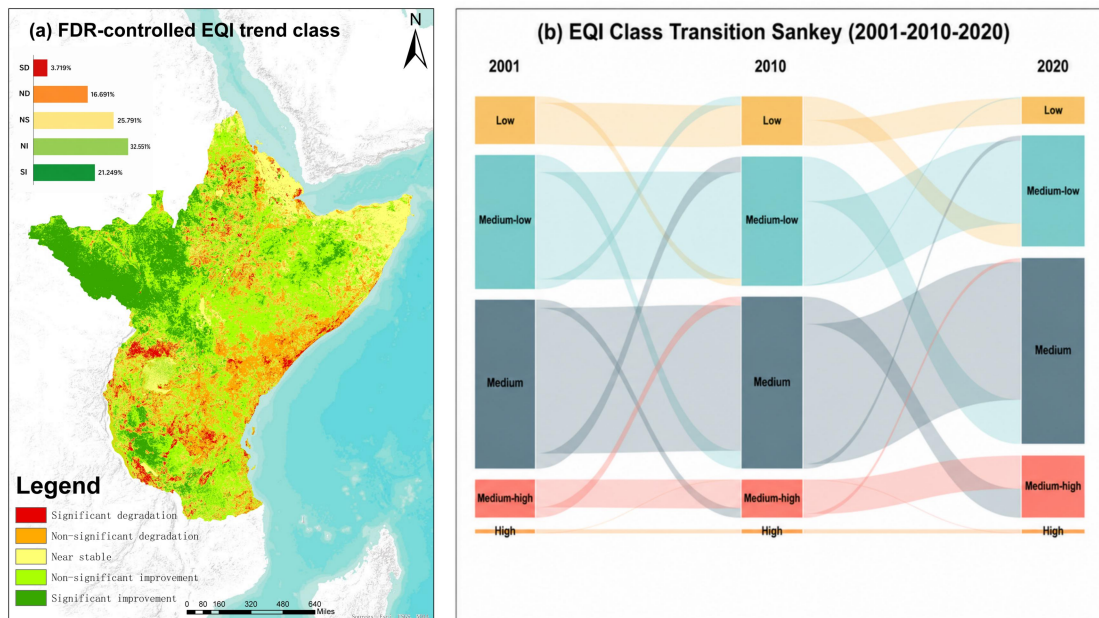


Figure 2 | FDR-controlled vegetation-based EQI trend classes and EQI class transitions. a, Pixel-wise FDR-controlled trend classes for 2001-2020. b, EQI class-transition Sankey diagram for 2001, 2010 and 2020. SD, significant degradation; ND, non-significant degradation; NS, near stable; NI, non-significant improvement; SI, significant improvement.

Land-cover context concentrates gains in open vegetation systems

Land-cover context clarified where regional vegetation-based EQI change accumulated. Grassland occupied 38.05% of the study area and made the largest area-weighted contribution to EQI change (+1.563 EQI units). Savanna and shrubland also contributed strongly (+0.897 and +0.673), reflecting their broad spatial extent and positive mean EQI changes across open vegetation belts (Extended Data Fig. 6). The land-cover-specific transition diagrams further show that these gains mostly occurred as shifts among neighbouring EQI classes rather than as wholesale conversion from low to high quality (Fig. 3c). These gains should be interpreted as increases in vegetation cover, canopy activity and productivity within open ecosystems, not as direct evidence of biodiversity recovery or ecosystem restoration.

The class-specific trajectories show persistent differences among land-cover contexts (Fig. 3a,b). Forest and cropland retained the highest vegetation-based EQI values, grassland and impervious mixed pixels occupied intermediate values, and water and barren classes remained low throughout the period. The 2001-2020 transition ribbons indicate that shrubland, water and impervious baseline contexts had relatively high shares of changed EQI classes, whereas cropland changed less (Fig. 3c). Water, wetland and impervious pixels were retained in the baseline maps to maintain a spatially exhaustive diagnosis, but they are not ecologically equivalent under a vegetation-based index. Masking baseline wetland, water and impervious pixels excluded only 3.17% of valid pixels and left significant improvement almost unchanged, while reducing significant degradation from 3.72% to 2.62% (Supplementary Table S7).





Figure 4 | *Climate-EQI co-variation used as descriptive context. a, Regional annual EQI, temperature, precipitation and wind-speed trajectories with Theil-Sen, Mann-Kendall and joinpoint diagnostics. b, Pairwise EQI-climate distributions and correlation matrix. These diagnostics motivate the model-based association analysis but are not treated as causal evidence.*

Space-time separated model associations

The modelling framework separated three ecological questions that are often pooled in a single modelling step. Model 1 quantified which covariates were associated with persistently high or low mean EQI; Model 2 quantified which climate trends, land-cover transition indicators and baseline modifiers were associated with long-term EQI slopes; and Model 3 quantified which annual anomalies were associated with a given location having EQI values above or below its own 2001-2020 mean. All three models used XGBoost regressors interpreted with SHAP values, but the predictor sets differed by design to avoid treating static constraints, dynamic trends and annual anomalies as interchangeable evidence.

Model 1 described the persistent spatial template of mean EQI. Soil_pH was the strongest spatial covariate, accounting for 31.3% of total mean absolute SHAP contribution, followed by Mean_Precip_mm (12.1%), LC_Shrubland (7.9%), LC_Grassland (6.9%) and Mean_Tmax_C (6.6%). These variables should be interpreted as long-term spatial covariates and ecological context that describe where mean vegetation-based EQI is high or low, not as temporal explanations for annual or decadal change.

Model 2 modelled spatial variation in long-term EQI slopes after removing baseline EQI and baseline FVC_norm, LAI_norm and GPP_norm, which would otherwise introduce circular model interpretation. The leading covariates were Dewpoint_C_Sen_slope (18.6%), Baseline_Soil_CFVO (10.4%), Baseline_LC_class (6.8%), Delta_LC_Forest (6.5%), WS10_ms_Sen_slope (6.1%), Temp_C_Sen_slope (6.1%), Precip_mm_Sen_slope (6.0%) and Tmin_C_Sen_slope (4.9%). This indicates that long-term vegetation-based EQI trends were most strongly associated with climate trends, edaphic background and land-cover transition context, rather than collapsing into a single dominant covariate.

Model 3 focused on within-location annual EQI anomalies and did not include engineered product interaction terms. Current-year precipitation anomaly was the strongest covariate (25.8%), followed by one-year lagged precipitation anomaly (12.3%), maximum-temperature anomaly (7.6%), mean-temperature anomaly (6.1%), lagged maximum-temperature anomaly (5.6%), lagged mean-temperature anomaly (5.5%), dewpoint anomaly (5.2%) and lagged dewpoint anomaly (4.5%). This result shows that interannual departures from local mean vegetation-based EQI were most closely associated with current and antecedent climate anomalies.

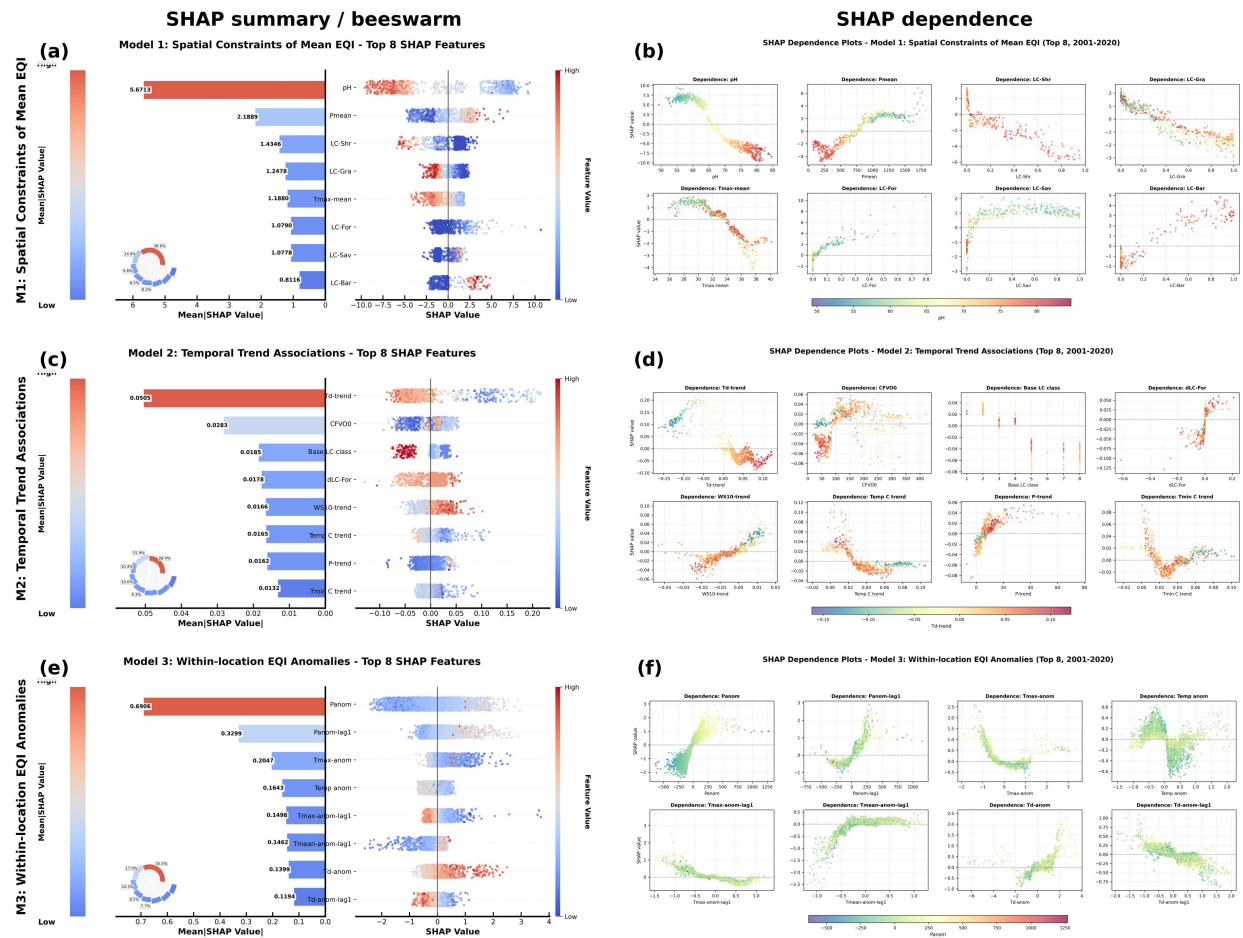


Figure 5 | Space-time separated SHAP model-interpretation diagnostics. The left column shows top-8 SHAP summary and beeswarm diagnostics, and the right column shows SHAP dependence plots. Rows correspond to Model 1 spatial covariates of mean EQI, Model 2 long-term trend associations and Model 3 within-location annual EQI anomalies. SHAP values summarize model-based associations and are not interpreted as causal effects.

Spatial validation and interpretation stability

Spatial block validation reduced apparent performance relative to random cross-validation, as expected for spatially autocorrelated environmental data, but the models retained useful predictive skill. Under 200 km spatial blocks, R2 values were 0.925 for Model 1, 0.675 for Model 2 and 0.590 for Model 3 (Supplementary Table S5). SHAP rankings also remained stable under blocking, with rank correlations of 0.968, 0.938 and 0.959 under 200 km blocks for Models 1-3, respectively (Supplementary Table S6). These results support the robustness of the model interpretations, not causal identification.

Land-cover coupling was tested directly by rerunning all three models without land-cover predictors. Removing land cover reduced 200 km block R2 from 0.925 to 0.882 in Model 1, confirming that land cover helps describe the persistent spatial template. Model 2 declined from 0.675 to 0.617, whereas Model 3 changed only from 0.590 to 0.586. The main climate-association patterns therefore did not depend on land-cover predictors, although land cover remains useful for contextual interpretation (Extended Data Fig. 3 and Fig. 4).

Discussion

The results point to a heterogeneous vegetation response rather than a simple regional greening story. This matters because recent global greenness assessments show that increasing satellite greenness can coexist with regional browning, tree-mortality pulses and slower recovery after disturbance [1] [3] [4]. In East Africa, the EQI signal was strongest in open vegetation systems and more localized in degradation zones,

which is consistent with landscapes where canopy cover is not saturated and small changes in water availability or productivity can produce large changes in vegetation indices [5] [11]. The component analysis supports the central ecological interpretation: FVC, LAI and GPP were broadly consistent, but GPP contributed most strongly to improvement hotspots, indicating that regional gains were primarily productivity-linked rather than simply denser vegetation cover.

The productivity-linked improvement is ecologically plausible in water-limited grassland, shrubland and savanna settings. African rangeland GPP is known to be tightly linked to precipitation, but the response depends on rainfall variability, vegetation type and local environmental setting [5]. The strong association with precipitation anomalies in the anomaly model therefore likely reflects pulse-like productivity responses after wet years or wetter seasons. The fact that lagged and contemporaneous climate anomalies both mattered is also consistent with dryland systems, where rainfall timing, intraseasonal redistribution and antecedent soil-water conditions can reshape productivity responses beyond the annual rainfall total [18] [19].

Atmospheric demand provides a plausible ecological interpretation for the co-occurrence of temperature, dewpoint and wind-related covariates with precipitation. Higher temperature can increase productivity where vegetation is below its thermal optimum, but it can also raise evaporative demand and weaken gains when water is limiting [10]. Recent evidence shows that terrestrial productivity is sensitive to surface wind speed through its influence on atmospheric dryness, stomatal conductance and soil drying [8]. Similarly, soil and atmospheric dryness can dominate vegetation-productivity responses at different thresholds, meaning that the same precipitation anomaly may have different effects depending on local edaphic and atmospheric background conditions [12]. This is consistent with our model separation: soil variables described the spatial template of mean EQI, whereas climate anomalies were most closely associated with year-to-year departures.

Land-cover context explains where EQI change accumulated, but it should not be read as an independent causal mechanism. Grassland, savanna and shrubland gains are consistent with open systems where vegetation cover and GPP respond quickly to rainfall pulses, grazing pressure, woody expansion or recovery from drought [5] [14]. Cropland-associated changes may reflect crop phenology, irrigation, fertilization or management intensification, all of which can raise vegetation indices without necessarily improving broader ecosystem condition [21] [22]. Water, wetland and impervious areas require even greater caution because a vegetation-based EQI is not designed to diagnose aquatic ecosystem quality, wetland function or urban ecological condition.

The spatial pattern of mean EQI suggests that persistent environmental constraints set the background on which temporal variability operates. Areas with favourable precipitation regimes and soil conditions maintained higher mean vegetation-based EQI, while arid and edaphically constrained zones were more likely to remain low or respond episodically. This is consistent with studies showing that productivity responses are shaped jointly by precipitation, temperature, atmospheric demand and soil moisture rather than by a single climate variable [5] [10] [12]. The implication is that static soil variables should be interpreted as spatial covariates, constraints or modifiers of vegetation potential, whereas precipitation, temperature and atmospheric-demand anomalies provide more direct model-based associations with interannual EQI deviations.

The main limitation is therefore conceptual rather than only technical. The EQI used here integrates FVC, LAI and GPP, so it diagnoses vegetation cover, canopy structure and productivity but cannot establish biodiversity recovery, habitat connectivity, aquatic condition or total ecosystem health [3] [24]. Internal sensitivity tests show that the main vegetation-based trend is not produced by one component or one normalization choice, but they do not replace field plots, high-resolution validation or process experiments. Within that boundary, the study identifies where vegetation-based ecological quality changed and supports a model-based ecological interpretation of climate-linked productivity gain in open vegetation systems, conditioned by edaphic and land-cover context.

Methods

Study area and datasets

The study area covers Kenya, Tanzania, Uganda, Rwanda, Burundi, Ethiopia, South Sudan, Somalia, Eritrea, Djibouti and the eastern Democratic Republic of the Congo. National boundaries were dissolved into one regional mask, and the analysis period was set to 2001-2020 to match complete annual MODIS

vegetation, productivity and land-cover records. All products were quality screened, clipped to the study area and harmonized to a 1 km grid in an equal-area projection; continuous variables were resampled bilinearly and categorical land-cover variables by modal class. Coordinates were retained only for mapping and spatial blocking.

Vegetation components were derived from MODIS: FVC from annual maximum MOD13A2 NDVI, LAI from MOD15A2H annual mean LAI and GPP from MOD17A3HGF annual productivity. MCD12Q1 land cover was reclassified into forest, shrubland, grassland, savanna, cropland, wetland, water, barren and impervious classes. Precipitation came from CHIRPS; temperature, dewpoint, wind and surface pressure from ERA5; soil variables from SoilGrids; and terrain from SRTM. Data sources and variable roles are listed in Supplementary Table S1.

Construction and robustness assessment of the vegetation-based EQI

We constructed annual vegetation-based EQI from 2001 to 2020 to represent three terrestrial vegetation dimensions: cover, canopy structure and productivity. MODIS-derived FVC_{norm} , LAI_{norm} and GPP_{norm} were used as the cover, structure and function components, respectively. The index is therefore a vegetation-based ecological-quality indicator, not a measure of biodiversity, aquatic condition or total ecosystem health.

Annual FVC_{norm} was derived from MOD13A2 annual maximum NDVI using the dimidiate pixel model:

$$FVC_{norm} = \text{clip} \left(\frac{NDVI - NDVI_{soil}}{NDVI_{veg} - NDVI_{soil}}, 0, 1 \right) \times 100 \quad (1)$$

where $NDVI_{soil} = 0.05$ and $NDVI_{veg} = 0.90$ were used as the default lower and upper vegetation-endmember thresholds. LAI and GPP were normalized to the same 0-100 scale as follows:

$$\begin{aligned} LAI_{norm} &= \text{clip} \left(\frac{LAI}{7}, 0, 1 \right) \times 100 \\ GPP_{norm} &= \text{clip} \left(\frac{GPP}{3}, 0, 1 \right) \times 100 \end{aligned} \quad (2)$$

The annual EQI was then calculated as the arithmetic mean of the three components:

$$EQI = \frac{FVC_{norm} + LAI_{norm} + GPP_{norm}}{3} \quad (3)$$

The equal-weight EQI was used as the baseline because the three components describe complementary vegetation dimensions rather than optimized causal weights. Internal robustness was assessed with four tests: component consistency between each component and default EQI; alternative component-weight schemes; normalization-threshold sensitivity for NDVI endmembers, LAI maximum and GPP maximum; and leave-one-component-out formulations. For FVC_{norm} threshold tests, FVC_{norm} was recomputed from annual NDVI before EQI construction.

For every alternative formulation, annual EQI, Theil-Sen slopes, Mann-Kendall p values, FDR-adjusted q values and trend classes were recalculated. Robustness was summarized using Spearman correlations for mean and slope maps and Jaccard overlap for significant improvement and degradation hotspots. These comparisons were interpreted as internal component robustness, not independent validation.

XGBoost-SHAP association framework

Model interpretation used XGBoost regression with TreeSHAP [31] [32]. XGBoost was chosen to capture nonlinear thresholds and interactions among climate, soil, terrain and land-cover variables, while SHAP values summarized both global importance, measured by mean absolute SHAP values, and feature-response patterns through dependence plots. We interpret these outputs as model-based associations rather than causal effects.

To avoid conflating persistent spatial gradients with temporal ecological change, the analysis was separated into three linked models with distinct response variables and interpretation targets.

Model 1 modelled mean EQI during 2001-2020 using soil, terrain, long-term mean climate and land-cover context, and was interpreted as a spatial constraint model. Model 2 modelled pixel-wise EQI Theil-Sen slopes using climate trends, land-cover change variables, baseline land-cover class, barren fraction and edaphic modifiers, while excluding baseline EQI and baseline FVC_{norm}, LAI_{norm} and GPP_{norm} to reduce circular model interpretation. Model 3 modelled within-location annual EQI anomalies, calculated as departures from each pixel's 2001-2020 mean, using contemporaneous and one-year lagged climate anomalies, land-cover change indicators and baseline context variables. Coordinates were excluded from all predictor sets.

Spatial block validation used regular equal-area grid blocks. Analysis-unit centroids were projected to Africa Albers. We used 100 km and 200 km blocks as empirical scale-sensitivity distances, following spatial cross-validation guidance that block size should reflect the spatial support of observations and plausible autocorrelation ranges when the true range is uncertain [33] [34] [35]. The 100 km blocks slightly exceeded the 80 km analysis unit and tested local spatial dependence; 200 km blocks provided a stricter regional transferability test. GroupKFold used block_id as the grouping variable, keeping all samples from the same block in the same fold. The 100 km and 200 km designs contained 484 and 141 valid blocks, respectively. For Model 3, all annual records from the same analysis unit and spatial block were kept together, preventing different years from the same location from entering both training and testing sets.

Trend analysis and significance testing

Temporal trends in EQI, EQI components and climate variables were estimated with Theil-Sen slopes and Mann-Kendall tests [26] [27] [28]. Theil-Sen slopes quantified monotonic rates of change and reduced sensitivity to outliers in annual remote-sensing series:

$$\beta = \text{median} \left(\frac{x_j - x_i}{j - i} \right), \quad 1 \leq i < j \leq n \quad (4)$$

where beta represents the annual rate of change. The Mann-Kendall test was used to evaluate trend significance [27] [28]:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sign}(y_j - y_i) \quad (5)$$

Pixel-wise trends were calculated for annual EQI and all robustness formulations. Mann-Kendall p values were adjusted with the Benjamini-Hochberg false-discovery-rate procedure [29]. Pixels were classified as significant improvement, non-significant improvement, near stable, non-significant degradation or significant degradation from the Theil-Sen slope and FDR-adjusted q value; near stable was defined as |Sen slope| < 0.05 EQI yr⁻¹.

A supplementary autocorrelation check calculated lag-1 autocorrelation from Theil-Sen-detrended EQI residuals. Where lag⁻¹ autocorrelation was significant, Mann-Kendall p values were recomputed using trend-free pre-whitening [30], followed by the same FDR correction and trend-class rules.

Regional mean EQI and climate series were evaluated with the same Theil-Sen and Mann-Kendall statistics. Joinpoint-style segmented fitting [36] was used only to visualize possible shifts in regional trajectories; spatial trend inference relied on Theil-Sen slopes, Mann-Kendall tests and FDR correction.

Data availability

The source datasets are publicly available from NASA LP DAAC, CHIRPS, ECMWF ERA5, SoilGrids, SRTM and Google Earth Engine data catalogues. Processed rasters, modelling tables, sensitivity outputs and figure source data are available from the corresponding author and will be deposited in a public repository before publication. No repository DOI has yet been assigned.

Code availability

Code used to fit the three XGBoost-SHAP models (spatial constraint, temporal trend association and within-location anomaly models) is provided in the accompanying archive East_Africa_EQI_model_code.zip. The archive contains the three model scripts, shared configuration, helper functions, requirements and a README

describing the required modelling tables. The full preprocessing workflow and code will be deposited in a public repository before publication; no repository DOI has yet been assigned.

Funding

This work was supported by the National Natural Sciences Fund Project (Nos. 42271097 and 42561144300) and the National Key Research and Development Project (No. 2022YFF1303204).

Author contributions

YuFeng Liu: led the project, wrote the manuscript, searched and curated the data, wrote the code and developed the modelling design. **Shiliang Liu:** acquired funding, contributed to the study ideas and revised the manuscript. **Qiang Zhang:** supervised. **Lam-Son Phan Tran:** supervised. **Bingkun Zang:** prepared the figures. **Xue Yan:** supervised. **Deresa Abetu Gadisa:** supervised. **John Mwangi:** supervised.

Competing interests

The authors declare no competing interests.

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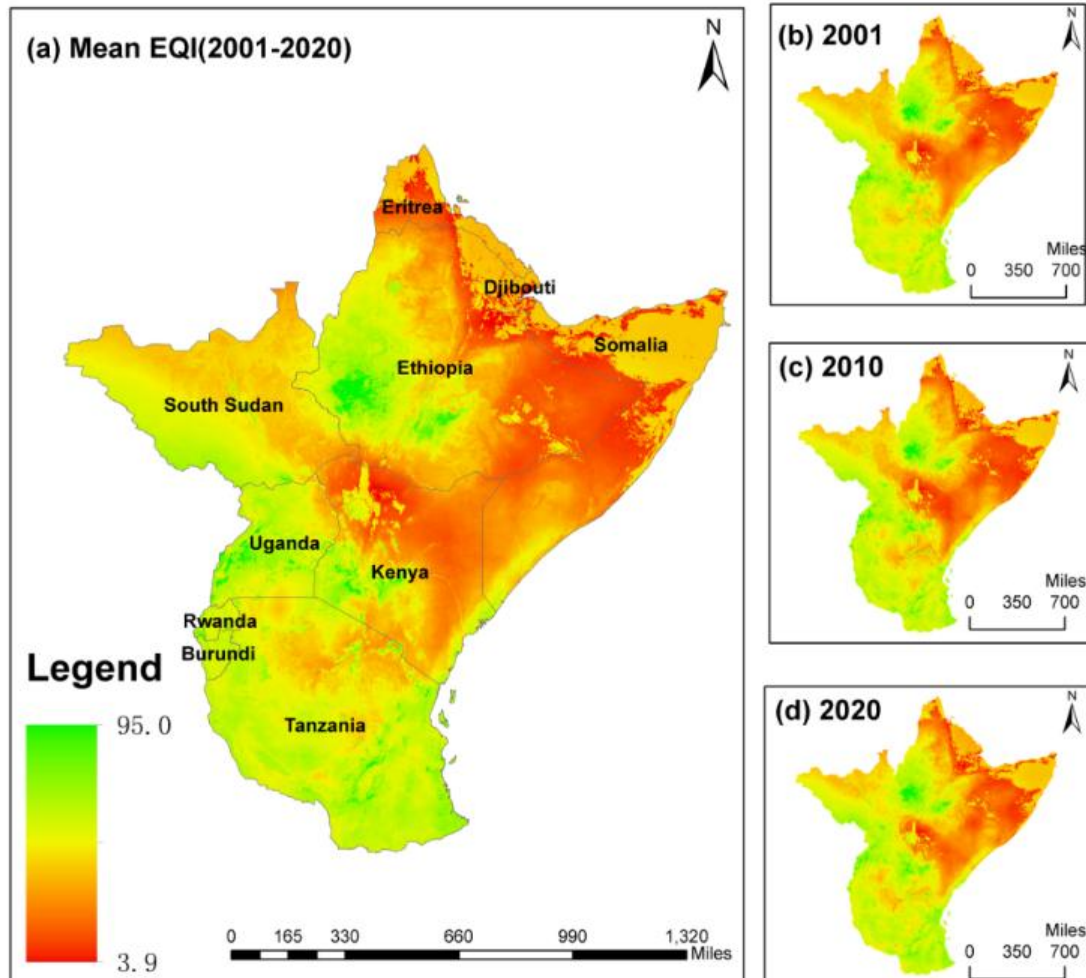
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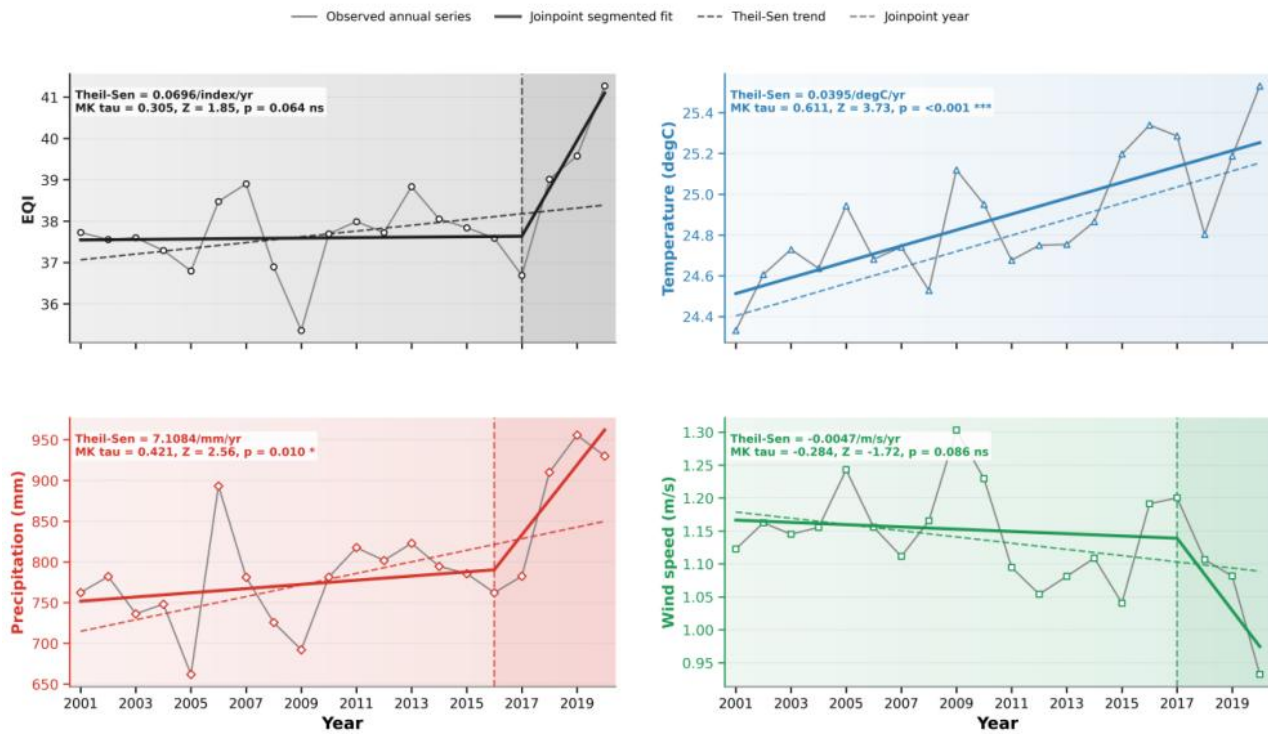
Extended Data Figures

Extended Data figures provide article back-matter evidence that supports the main text. They are separated from the Supplementary Information, which is reserved for definitions, equations, notation and methodological details.

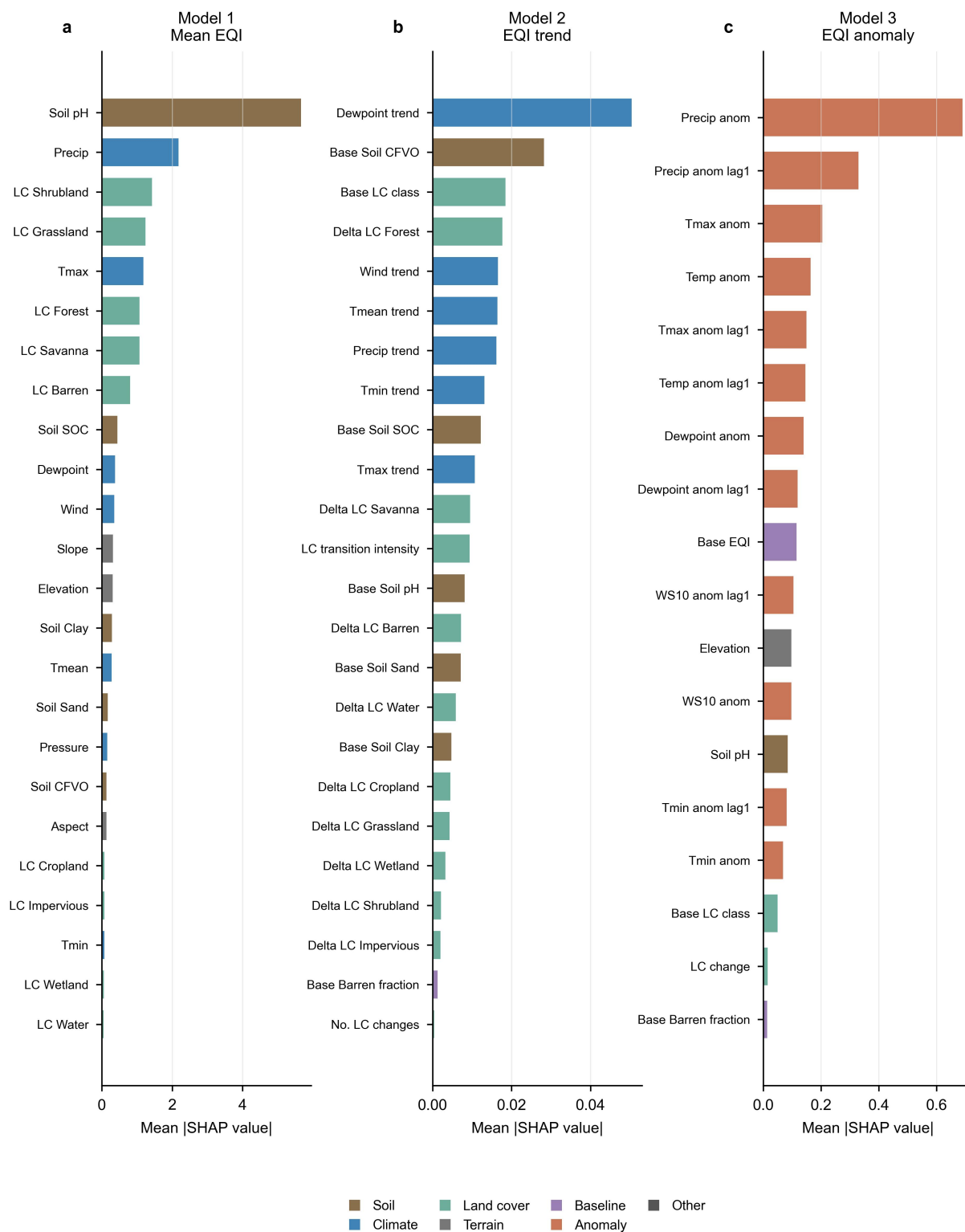


Extended Data Fig. 1 | Spatial pattern and temporal snapshots of vegetation-based EQI. The left panel shows the 2001-2020 mean EQI across East Africa, and the right panels show annual snapshots for 2001, 2010 and 2020. The map supports the spatial-template interpretation in the main text by showing persistent humid-highland and lake-region maxima and dryland minima. EQI represents vegetation cover, canopy structure and productivity, not total ecosystem condition.

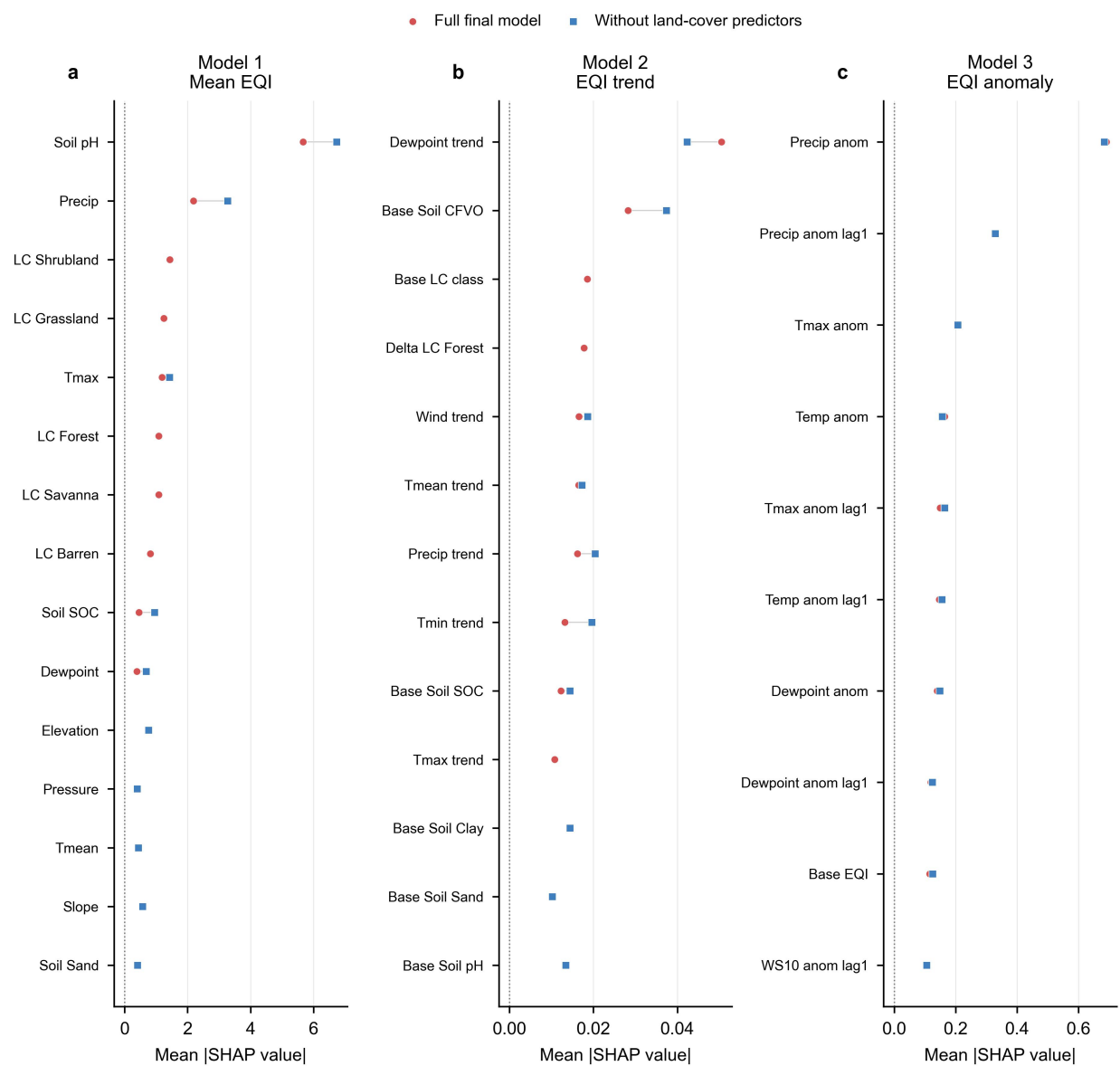
EQI and Climate Features with Theil-Sen, Mann-Kendall and Joinpoint Diagnostics (2001-2020)



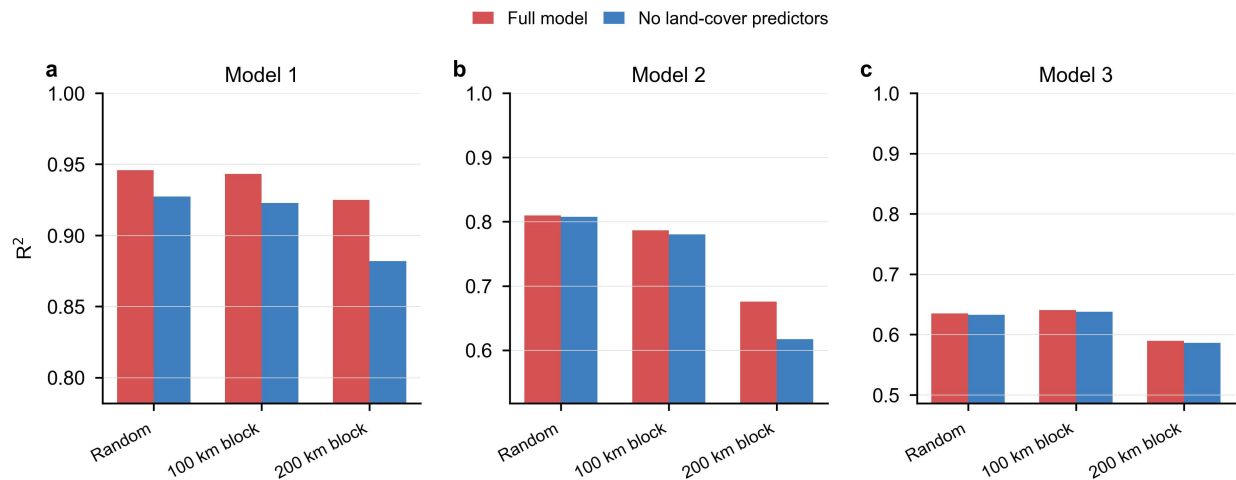
Extended Data Fig. 2 | Regional EQI and climate time-series diagnostics. Annual regional EQI, temperature, precipitation and wind-speed series are shown with Theil-Sen slopes, Mann-Kendall diagnostics and joinpoint fits for 2001-2020. These co-variation diagnostics provide descriptive temporal context for the association models but are not interpreted as causal evidence.



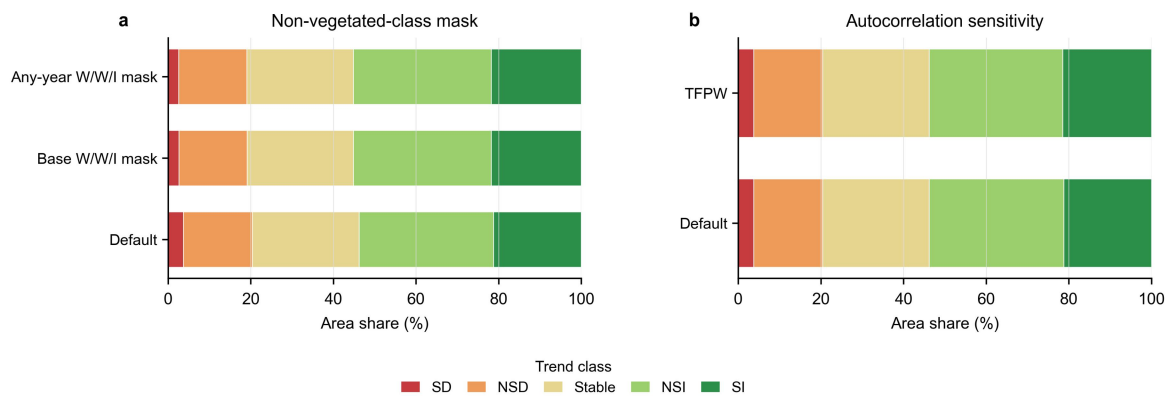
Extended Data Fig. 3 | Full SHAP rankings for the three association models. All retained predictors are shown for the spatial constraint model, temporal trend model and within-location anomaly model. Colours indicate predictor families and mean absolute SHAP values quantify relative model contribution strength.



Extended Data Fig. 4 | SHAP rankings after removing land-cover predictors. Points compare full final models with sensitivity models in which land-cover predictors were excluded. This tests whether the major model-interpretation rankings are dominated by land-cover context variables.



Extended Data Fig. 5 | Predictive performance after removing land-cover predictors. Random and spatial block cross-validation R² values are compared between full models and sensitivity models without land-cover predictors.



Extended Data Fig. 6 | Trend-class sensitivity to non-vegetated masking and trend-free pre-whitening. Stacked bars compare FDR-controlled trend-class area shares after masking wetland, water and impervious pixels and after correcting for temporal autocorrelation using trend-free pre-whitening. SD, significant degradation; NSD, non-significant degradation; Stable, near stable; NSI, non-significant improvement; SI, significant improvement. W/W/I, wetland/water/impervious; TFPW, trend-free pre-whitening.